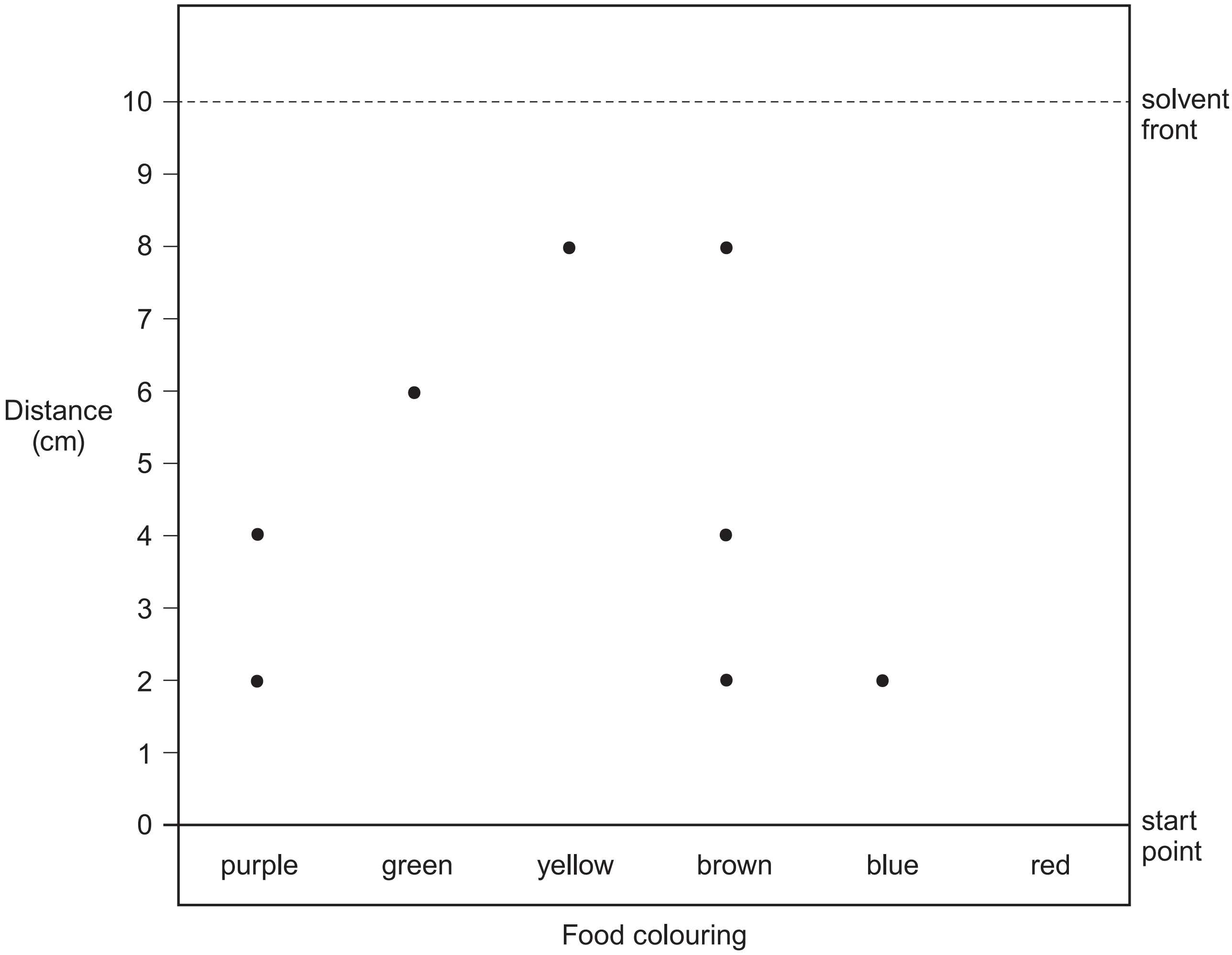
5. Analytical scientists work in a wide range of different industries and agencies.

(a) The Food Standards Agency (FSA) commissioned a project to detect the coloured dyes used in food colourings.

The diagram below shows a chromatogram of different food colourings.



Examiner only

(i) The result for the dye in the red food colouring is not included in the diagram. The R_f value for red colouring is 0.4.

Use the equation:

$$R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by solvent}}$$

to calculate the distance moved by the dye in the red colouring on this chromatogram **and** add it to the diagram. [3]

distance travelled = ... cm

(ii) It was claimed that brown colouring was not a combination of other coloured dyes. Explain whether you agree with this claim. [2]

....

....

....

(iii) Explain, in terms of the mobile and stationary phases, why the molecules in purple colouring move different distances along the chromatogram. [3]

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Examiner only

(b) Qualitative chemical testing is used to identify the ions in compounds.

(i) Six unlabelled bottles each contained a solution of one of the compounds in the box below.

copper(II) sulfate	sodium carbonate	lithium carbonate
sodium chloride	potassium sulfate	calcium iodide

I. State which compounds would produce a yellow flame in a flame test. [1]

....

II. State which compounds would produce bubbles of gas when added to hydrochloric acid. [1]

....

(ii) A technician has a colourless solution in an unlabelled bottle. He suspects it is metal chloride. Describe a test that could be carried out to confirm that the solution is a chloride, and give the expected result. [3]

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